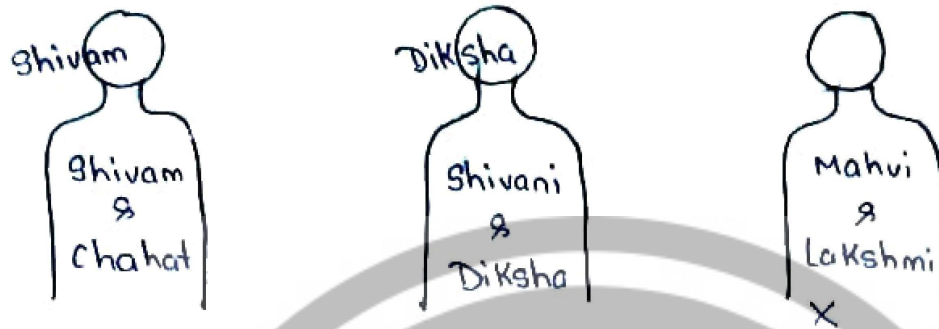


SETS

Two best players in Indian Cricket team
→ not well defined



well defined ~~नहीं है~~

Natural Numbers less than 10

1, 2, 3, 4, 5, 6, 7, 8, 9

Collection of vowel of English alphabet
a, e, i, o, u

well defined
Collection

Two best Actresses → well defined
~~नहीं है~~ X

Jo person to person
change na ho

Sets

A well-defined collection of objects or Element is called a Set.

Example

① Even Natural Number less than 15

2, 4, 6, 8, 10, 12, 14

② Prime Natural Number less than 10

2, 3, 5, 7

object
or
Element

↓ ↓ ↓ ↓

object object object object.

Notation of Sets

Sets are mainly denoted by capital letters of the English alphabet like - A, B, C X, Y, Z etc

$$A = \{ \quad \}$$

$$B = \{ \quad \}$$

$$C = \{ \quad \}$$

Elements of Sets

The objects stored in a set are called elements. They are denoted by small letters of the English alphabet.

Like: a, b, c x, y, z etc.

$$\{ \dots \}$$

Representation of Sets

Sets are represented in two ways

1. Tabular or Roster method: Elements are listed using commas and brackets

Example

A = Collection of Even Natural Number less than 10

= 2, 4, 6, 8 → Element

↓ ↓ ↓
Element Element Element

$$A = \{ 2, 4, 6, 8 \}$$

2. Construction or property method: In this form, instead of listing all the elements, we write the general rule satisfied by the elements of the set.

Example

$$A = \{ 2, 4, 6, 8 \dots \}$$

$A = \{x : x \text{ is an Even Natural Number}\}$
→ such that

$A = \{x | x \text{ is an Even Natural Number}\}$
→ such that

Roster form

Set Builder form

Note

- if 'a' is an Element of set A, then it is represented by $a \in A$,
- if 'a' is not an Element of set A, then it is represented by $a \notin A$ (does not belong to)
- Elements of sets are not Repeated $\rightarrow \{2, 2, 3, 3\} \Rightarrow \{2, 3, 1\}$
- Order of Set is Immaterial
 $\begin{matrix} \rightarrow \{3, 2, 1\} \\ \rightarrow \{3, 1, 2\} \\ \rightarrow \{2, 1, 3\} \end{matrix}$

Types of Sets

- 1) Null Set
- 2) Singleton Set
- 3) Finite Set
- 4) Infinite Set
- 5) Equal Set
- 6) Equivalent Set.

Null Set or Empty Set or Void Set

A Set which does not contain any Element is called Empty Set or null set or void set. and it is denoted by $\emptyset$ or $\{ \}$

Example

(I) collection of Intersecting points of two parallel lines

(II) $A = \{x : x \text{ is a natural number } \& \ 5 < x < 6\}$

$A = \{ \}$

Singleton set

A Set Containing Single Element is called Singleton Set.

Example

(I) $A = \{x : x \text{ is an Even prime Number}\}$

$$A = \{2\}$$

(II) $B = \{x : x \text{ is prime Number, } 3 < x < 7\}$

$$B = \{5\}$$

Finite set

A Set which is either Empty or has finite number of Elements is called a finite Set.

Example

$A = \{x : x \text{ is an Even natural no. less than } 100\}$

$$A = \{2, 4, 6, \dots, 98\}$$

$$B = \{1, 3, 5, 7, 9\}$$

$$C = \{2, 5, 7, 11, 13, 3\}$$

Note:

Number of Elements in a finite Set A is called the Order or the Cardinality of the Set A, and it is denoted by $o(A)$ or $n(A)$

$$A = \{3, 2, 1, 4\}$$

$$n(A) = o(A) = 4$$

$$B = \{1, 3, 5, 7, 9, 11, 13, 15\}$$

$$n(B) = 8$$

(III)

$$C = \{\}$$

$$n(C) = 0$$

(IV)

$$D = \{2\}$$

$$n(D) = 1$$

Infinite Set

A Set having Infinite number of Elements is called infinite Set.

Example

(i) $A = \{x : x \text{ is an Even Natural Number}\}$

$$A = \{2, 4, 6, 8, \dots\}$$

(ii) $B = \{x : x \text{ is a prime Natural Number}\}$

$$B = \{2, 3, 5, 7, 11, \dots\}$$

Equal Set

Two Set are said to be Equal if they have exactly the same Elements.

Example

(i) $A = \{a, b, c, d\}$ $B = \{b, c, a, d\}$

$$\boxed{A = B}$$

(ii) $A = \{1, 2, 3, 4\}$ $B = \{2, 1, 3, 4\}$ $C = \{3, 4, 1, 2\}$

$$\boxed{A = B = C}$$

Equivalent Sets

Two finite sets are said to be Equivalent if their Cardinalities are Equal.

Example

(i) $A = \{1, 2, 3, 4\}$ $B = \{a, b, c, d\}$

$$n(A) = 4$$

$$n(B) = 4$$

$$\boxed{n(A) = n(B)}$$

$\therefore A \& B$ are Equivalent Sets

Note :-

All Equal Set are Equivalent sets.

subset

Two Sets A and B in which all the elements of A are elements of B, but it is not necessary that the elements of B are elements of A. Then A is called a subset of B, and it is represented by $A \subset B$.

Subset
According to N.C.E.R.T.
 $A \subset B$

Example $\Rightarrow$ Let $A = \{1, 2, 3\}$ $B = \{1, 2, 3, 4, 5\}$

$\therefore$ all elements of Set A are elements of Set B
 $\therefore$ A is a subset of B.

$$A \subset B$$

Example

$A = \{a, b, c, d\}$ $B = \{b, a, c, d\}$ $A = B$

$\therefore$ all elements of Set A are elements of Set B
 $\therefore$ A is a subset of B

$$A \subset B$$

$\therefore$ all elements of Set B are elements of Set A
 $\therefore$ B is a subset of A
 $B \subset A$

Note $\Rightarrow$ $A = B \Rightarrow A \subset B$ & $B \subset A$

Note $\Rightarrow$

(I) Empty Set is always a subset of every set

(II) Every set is a subset of itself.

(III) If no. of elements in a set A is n then total number of subsets in set A = 2^n

Example $\Rightarrow A = \{1, 2, 3, 4\}$ Total no. of Subset.

$$n = 4$$

$$\text{Total no. of Subset} = 2^n$$

$$= 2^4$$

$$= 16 \quad \underline{\text{Ans}}$$

Qus II $A = \{a, b, c\}$ Total no. of Subset

$$n = 3$$

$$\text{Total no. of Subset} = 2^n = 2^3$$

$$= 8$$

Subset $\Rightarrow \phi, \{a\}, \{b\}, \{c\}, \{ab\}, \{ac\}, \{b, c\}, \{a, b, c\}$

Example $A = \{1, 2\}$

Subset $\Rightarrow \phi, \{1\}, \{2\}, \{1, 2\}$

Proper Subsets

If two sets A and B have all the elements of A in which B has at least one element more, then A is called a proper subset of B.

$$A \subset B \quad A \neq B$$

Interval

There are three types of intervals in a set of real numbers if R is a set of real numbers where a and b are elements of R but $a < b$

$$x \in R \quad a < b$$

(I) Open Interval

(II) Closed Interval

(III) Semi open & Semi closed Interval

Open Interval

$$a < x < b, x \in \mathbb{R}$$



$$x \in (a, b)$$

$$\text{or } x \in]a, b[$$

open Interval

Closed Interval

$$a \leq x \leq b$$



$$x \in [a, b]$$

closed Interval

1.001 $\nearrow$
 $x < 2$
1.9999 $\nwarrow$
 $1 \leq x \leq 2$

Semi Closed Interval / Semi open Interval

$$a \leq x < b$$



$$x \in [a, b)$$

$$a < x \leq b$$



$$x \in (a, b]$$

Universal Set

The main set which has other possible subsets is called the Universal Set, it is usually denoted by U

Power Set

The Set of Subsets of a Set is called the power Set. The power set of a set A is often denoted by $P(A)$.

Ex. $A = \{1, 2, 3\}$

Subset: $\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$

$$P(A) = \text{power set} = \{\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

Venn Diagram of Sets

→ Sets को diagram से Represent करेंगे

Universal Set $\Rightarrow$ Rectangle से Represent करते हैं

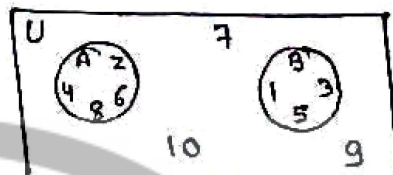
Subsets $\Rightarrow$ Circle से Represent करेंगे

Example

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{2, 4, 6, 8\}$$

$$B = \{1, 3, 5\}$$



Union of Sets $\Rightarrow$

$$A = \{2, 4, 6, 8\} \quad B = \{1, 3, 5\}$$

$$A \cup B = \{2, 4, 6, 8, 1, 3\}$$

→ union

$$* A = \{1, 2, 3, 4, 5, 6, 7\} \quad B = \{2, 5, 7, 10\} \quad C = \{3, 6, 9, 12\}$$

$$A \cup B \cup C = \{1, 2, 3, 4, 5, 6, 7, 10, 9, 12\}$$

Union of Sets

Union of any two sets A and B is a set whose all elements are either in A or in B or in both A & B.

Example

$$A = \{1, 3, 5, 7, 9\} \quad B = \{2, 4, 6, 8, 10\}$$

$$A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

U $\rightarrow$ Universal Set

U $\rightarrow$ Union

Properties of Union of Sets

- Commutative law $\Rightarrow A \cup B = B \cup A$
- Associative law $(A \cup B) \cup C = A \cup (B \cup C)$
- Identity law $A \cup \phi = \phi \cup A = A$ $\phi \rightarrow$ Identity element
- Idempotent law $A \cup A = A$ $\phi \rightarrow$ Null set

Intersection of Sets

The intersection of any two sets A and B is the set whose all elements are in both A and B.

Example * $A = \{2, 4, 6, 8, 10\}$ $B = \{2, 3, 5, 7\}$

$$A \cap B = \{2\}$$

↳ Intersection.

* $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$ $B = \{2, 4, 5, 6, 7\}$ $C = \{2, 3, 5, 7, 11\}$

$$A \cap B \cap C = \{2, 5, 7\}$$

Properties of Intersection of Sets

- Commutative law $A \cap B = B \cap A$
- Associative law $(A \cap B) \cap C = A \cap (B \cap C)$
- ϕ & U Rule $\Rightarrow A \cap \phi = \phi \cap A = \phi$ $U \cap A = A$
- Idempotent law $A \cap A = A$
- Distribution law $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

$$U = \{1 \text{ --- } 10\}$$

$$A = \{2, 3, 4\}$$

$$U \cap A = \{2, 3, 4\} \\ = A$$

Disjoint Sets

if both sets A and B have no element in common then they are called disjoint sets.

Example (I) $A = \{2, 4, 6, 8, \dots\}$

$$B = \{1, 3, 5, 7, \dots\}$$

$$A \cap B = \{ \} = \phi$$

$$\therefore A \cap B = \phi$$

$\therefore A$ & B are called disjoint set

(II) $A = \{2, 4, 6, 8, \dots\}$

$$B = \{2, 3, 5, 7, \dots\}$$

$$A \cap B = \{2\}$$

$$\therefore A \cap B \neq \phi$$

$\therefore A$ & B are not disjoint set

Difference of Sets

The difference $A-B$ of two sets A and B is the set of those elements which are in A but not in B .

$A-B = A$ से B के element हटा दो

$B-A = B$ से A के element हटा दो

Example $A = \{2, 3, 4, 5, 6\}$ $B = \{5, 6, 7, 8, 9\}$

$$A-B = \{2, 3, 4\}$$

$$B-A = \{7, 8, 9\}$$

Complement of a Set

Let U is an Universal Set and A is a subset of U then $(U-A)$ is called complement of a set A . and it is represented by A' or A^c

i.e. $A' = A^c = U-A$

Example $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ $A = \{2, 4, 6, 8, 10\}$ then find A' or A^c

$$A^c = A' = U-A$$

$$A^c = A' = \{1, 3, 5, 7, 9\}$$

Some Properties of Complement of a Set

(I) $(A')' = A$

(II) $A \cap A' = \phi$

(III) $A \cup A' = U$

(IV) De Morgan's law

(a) $(A \cup B)' = A' \cap B'$

(b) $(A \cap B)' = A' \cup B'$

(V) $\phi' = U - \phi$

$$\phi' = U$$

(VI) $U' = U - U = \phi$

$$U' = \phi$$